HARDWARE

Week 2

OBJECTIVES

1. Understand the function of the hardware components of a computer system.
2. Understand the function of different types of main memory.
3. Understand the concept of a stored program and the role of components of the CPU.
4. Understand factors that affect the performance of the CPU.

VON NEUMANN ARCHITECTURE

A computer system design in which the processing instructions are stored in memory with the data is called a stored-program computer.

COMPONENTS

1. Central Processing Unit (CPU):

Hardware device that carries out the processing in a computer.

2. Main Memory/Random-access Memory (RAM):

A temporary store for data and instructions (programs)

3. Bus

A group of connections between devices in a computer.

FETCH-DECODE-EXECUTE CYCLE

- Sequence of steps carried out repeatedly by a CPU

When a computer is turned on, the CPU **fetches** an instruction from the memory, carries out the instruction it receives, and then fetches the next, and so on.

INPUT & OUTPUT (I/O) DEVICES

- Input devices such as keyboards, mice and light sensors provide the computer with data it can process.
- Output devices such as monitors, printers and speakers are used by the computer to communicate the results of the processing.

SUBJECT VOCABULARY

MEMORY ADDRESS

a number that uniquely identifies a (memory) storage location.

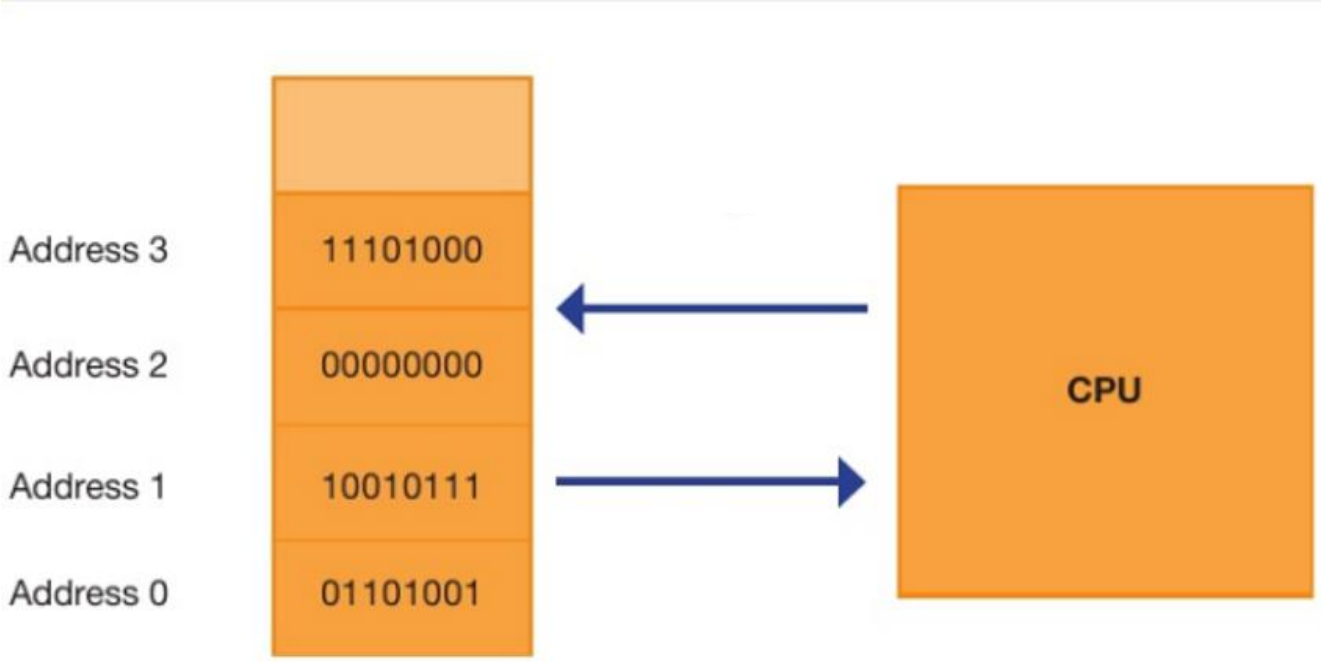
WRITING

When the CPU sends data to memory to be stored at a given address.

READING

When the CPU retrieves the data stored at a given address.

EXAMPLE


Address 3
11101000
Address 2
00000000
Address 1
10010111
Address 0
01101001
CPU

RAM VS ROM

- RAM is described as volatile, which means that its contents are lost when the power is turned off.
- Because of this, computers also need non-volatile memory to store any programs that must run when the computer is first turned on. this memory is called **read-only memory (ROM).**
- These programs are small and carry out specific tasks, such as initializing the hardware components and starting the operating system when a computer is switched on.

CACHE MEMORY

memory used to make up for the difference in speed between two internal components (CPU & RAM).

CHECK PAGES (165 – 168)